

Glutamate measurement

Material:

1. 22G needle
2. 1 ml syringe
3. Cold chamber
4. Cerebrospinal fluid sample
5. NADP⁺ (final concentration 1.0 mM)
6. CaCl₂ (final concentration 1.0 mM)
7. Glutamate dehydrogenase (50 units)
8. Glutamate (final concentration 1 mM)

Procedures:

1. Immediately after euthanasia, the rat head is flexed downward at approximately 45°, to reveal a palpable depressible surface with the appearance of a rhomb between the occipital protuberances and the spine of the atlas.
2. The 22G needle is inserted in the cisterna magna for cerebrospinal fluid (CSF) collection and connected to a 1-mL syringe. Then, 50-80 µL of non-contaminated sample is drawn into the syringe by simple and careful aspiration, and maintained in a refrigerator -80°C.
3. Transfer the CSF to the housing of the fluorimeter, and fluorescence at 450 nm is recorded after excitation at 360 nm. Then immediately, add NADP⁺ (final concentration 1.0 mM), follow by CaCl₂ (final concentration 1.0 mM).
4. After a further 70s period, add glutamate dehydrogenase (50 units) and monitor NADPH⁺ fluorescence allowed until reaching the equilibrium state, visualizing as a constant fluorescence value, made to measure the glutamate in different samples (5µL).
5. Subsequently, after 500s period, perform the calibration by adding standard glutamate (final concentration 1mM) to the system.
6. Results are described as nmol/µL of CSF to the maximum response obtained with the final addition of 1mM of glutamate.

References:

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